Polynomial-time simulation of non-Clifford quantum error correction

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Joint work with



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Overview

Most magic-state preparation protocols can be simulated in polynomial-time

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1 **Most magic-state preparation protocols**

Circuits made of CNOT, T, Paulis, Z measurements and *compatible* X measurements.

Examples:

- Magic state distillation and cultivation based on applying a logical 3rd level non-Clifford gate to a code.
- Code switching between 3D and 2D codes.

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

1. Most magic-state preparation protocols
2. **Simulated**

Maintain state representation throughout the circuit.

Can compute the following:

- Pauli expectation values
- Measurement samples
- Amplitudes
- State equality

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

1. Most magic-state preparation protocols
2. Simulated
3. **Polynomial time**

Polynomial-time complexity in the number of qubits and circuit depth
No exponential in the number of T gates!

Outline

1. The phase polynomial representation

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New question: what is $U |x\rangle$ for $U \in G_3$?

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$$T_i\,|x\rangle = \omega^{x_i}\,|x\rangle$$

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for a polynomial

$$f(x) = \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j + \sum_{i<j<k} c_{ijk} x_i x_j x_k$$